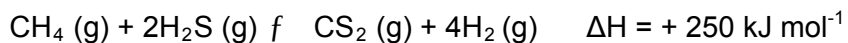


- 1 (a) Hydrogen sulfide is a common contaminant in many of the world's natural gas (NG) wells. Approximately one-third of the NG resources worldwide can be considered as low or sub-quality gas unfit for pipeline shipment. Presently in NG processing hydrogen sulfide is removed by method that is similar to the steam reforming of methane, one of which is represented by the equation below:



In a sealed 50 m³ container at 800 °C, methane and hydrogen sulfide were mixed in a 1:2 molar ratio. The pressure in the container increased from 7.5 atm to 10 atm.

- (i) Find the partial pressures of the four gases at equilibrium.

[3]

- (ii) Define the term K_p using the above equilibrium as an example.

- (iii) Hence calculate the K_p value at 800 °C.

[2]

[Turn over]

1 (a) (iv) Explain how the above equilibrium and K_p are affected by

(I) decrease in temperature.

.....

.....

.....

.....

(II) allowing the sealed container to expand.

.....

.....

.....

.....

[4]

(b) A solution of ammonium chloride is acidic.

(i) Given that the K_b of ammonia is $1.8 \times 10^{-5} \text{ mol dm}^{-3}$, calculate the pH of a $0.100 \text{ mol dm}^{-3}$ ammonium chloride solution.

[2]

[Turn over

- 1 (b) A buffer solution is produced by mixing 40.0 cm^3 of 0.1 mol dm^{-3} ammonium chloride solution with a certain volume of 0.2 mol dm^{-3} sodium hydroxide.
- (ii) Calculate the volume of 0.2 mol dm^{-3} sodium hydroxide required to create a buffer of maximum buffer capacity. Hence calculate the pH of the resulting buffer solution.

[2]

[total: 13]

- 2 (a) Pepsin C is an enzyme that is released in the stomach and degrades food proteins into peptides. It was the first animal enzyme to be discovered. The optimum working conditions for pepsin C are at pH of 1.5 – 2.0 and at temperatures of 37 – 42 °C.

- (i) What is understood by the term **primary structure** of protein?

.....
.....

[1]

- 2 (a) (ii) Partial hydrolysis of a polypeptide containing **6 amino acid residues** in pepsin C yields the following fragments.

Asn-Tyr, Glu-Asn-Tyr, Gly-Asp-Glu, Ile-Gly

Deduce the structure of this polypeptide.

.....

.....

.....

.....

[1]

- (b) Some of the most common amino acids in pepsin C are shown in **Table 1**:

Table 1:

Amino acid	Formula of side chain R of amino acid $\text{NH}_2\text{-CHR-COOH}$
Valine	$-\text{CH}(\text{CH}_3)_2$
Serine	$-\text{CH}_2\text{OH}$
Glutamic acid	$-\text{CH}_2\text{CH}_2\text{COOH}$
Lysine	$-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$
Aspartic acid	$-\text{CH}_2\text{COOH}$
Phenylalanine	$-\text{CH}_2\text{C}_6\text{H}_5$
Cysteine	$-\text{CH}_2\text{SH}$

- (i) The cysteine residues in pepsin C can form disulfide bridges. The formation of disulfide bonds is an oxidation process. Illustrate the formation of disulfide bridges between two cysteine molecules by means of a balanced equation.

.....

.....

[1]

- 2 (b) (ii) Pepsin C is a globular protein and the globular shape is held in place by various side-chain interactions such as disulphide bridges. Describe three other types of side-chain interactions; illustrate your answers with suitable pairs of amino acids from the **Table 1**.

.....

.....

.....

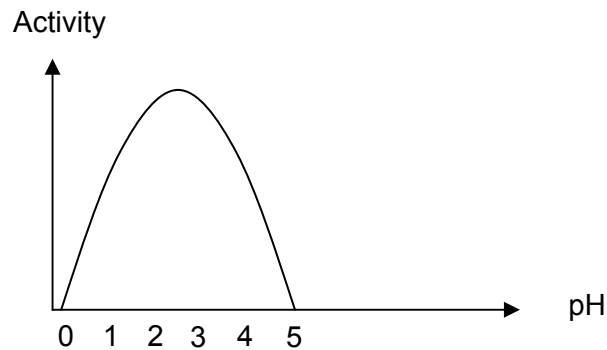
.....

.....

.....

[3]

- (c) Some samples of pepsin C were extracted from the body and were tested for its activity under different experimental conditions. The graph below shows the variation of the activity of the digestive enzyme when subjected to different pH:



- (i) Explain the activity of the digestive enzyme at pH 0 and pH 5.

.....

.....

.....

.....

[2]

- 2 (c) (ii) Suggest another factor that will cause pepsin to lose its activity.

.....

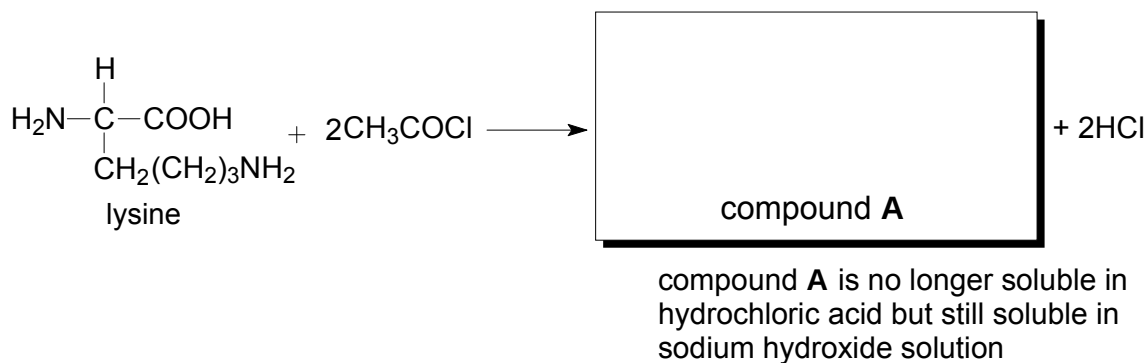
.....

.....

.....

[2]

- (d) Lysine is an amino acid found in pepsin C which is soluble in both acid and alkali. Consider the following reaction scheme:



- (i) Draw the structure of compound A in the box provided.
- (ii) Explain the difference in the solubility of lysine and compound A in acid and alkali.

.....

.....

.....

.....

.....

.....

.....

[4]

[total: 14]

- 3 Aluminium is the most abundant metal in the Earth's crust. It is a useful metal that can be extracted from bauxite ore by electrolysis. It exhibits amphoteric properties due to the presence of both ionic and covalent character in aluminium oxide compound.

(a) (i) Illustrate the amphoteric character of Al_2O_3 with the use of relevant equations.

.....

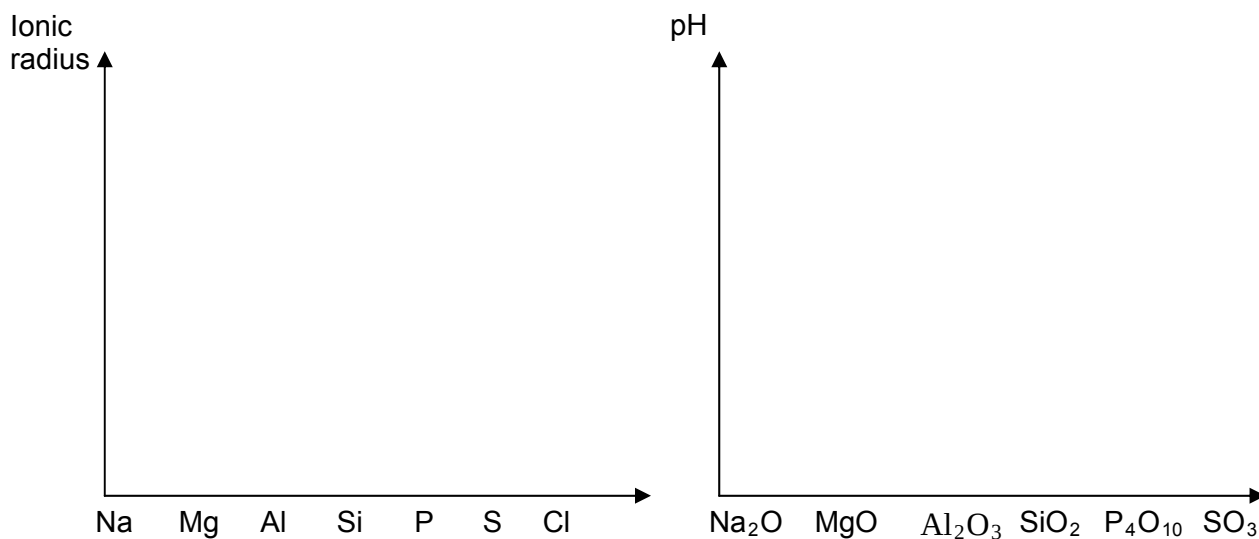
.....

.....

.....

[2]

(b) (i) Aluminium belongs to one of the elements in Period 3. Sketch the ionic radius of the elements in period 3 and the pH of the solution produced when the highest oxide of these elements are added to water separately.



[Turn over

- 3 (b) (ii) Explain the shape of your sketches.

Ionic radius:

.....

.....

.....

.....

.....

pH:

.....

.....

.....

.....

.....

.....

[6]

- (c) The period 3 metals undergo vigorous reactions with dry chlorine on heating to form their respective chlorides. One example of a chloride formed is MgCl_2 .

- (i) The dissolution of MgCl_2 produces a solution of pH 6.5. Explain this observation with the use of a relevant equation.

.....

.....

.....

[1]

- 3 (c) (ii) Construct a Born-Haber cycle in the grid below to calculate the enthalpy change of formation for MgCl_3 . Use the relevant data given below as well as from the *Data Booklet*.

Enthalpy change of atomization of Mg	+146 kJ mol^{-1}
1st electron affinity of Cl	-364 kJ mol^{-1}
$\Delta H_f (\text{MgCl}_2)$	-652 kJ mol^{-1}
Lattice energy of $\text{MgCl}_3 (\text{s})$	-5010 kJ mol^{-1}

Energy / kJ mol^{-1}



[4]

- 3 (c) (iii) Using thermodynamic concepts and the data in c(ii), explain why the formation of MgCl_3 is not as feasible as the formation of MgCl_2 under standard conditions.

.....

.....

.....

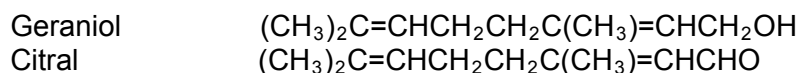
.....

.....

[3]

[total: 16]

- 4 Many of the natural compounds which are responsible for aromas in perfumes were first extracted from plants. Geraniol is found in roses and citral in lemon grass.



Both of these are attractants for honey bees whereas $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{OCOCH}_3$, 3-methylbutylethanoate is a bee alarm pheromone and signifies danger to a honey bee but is also the principal cause of smell of bananas.

- (a) (i) Suggest the reagents and conditions needed to convert geraniol into citral in the laboratory.

.....

[1]

- (ii) Describe a chemical test you could do to check that geraniol has been converted into citral.

.....

.....

.....

.....

[2]

[Turn over]

- 4 (a) (iii) Citral can react with hydrogen cyanide to give a product mixture.

By making specific reference to the slow step of the mechanism, explain why the product mixture obtained does not exhibit optical activity even though the product has a chiral centre.

.....

.....

.....

.....

.....

.....

[3]

- (b) (i) If 3-methylbutylethanoate is warmed with aqueous sodium hydroxide solution, a slow reaction takes place to produce sodium ethanoate and another product, **B**.

Write a chemical equation to represent this reaction giving the structure of product **B**.

.....

[1]

4 (b) (ii) Give a brief description to suggest how the kinetics of this reaction can be followed. Your answer should show clearly

(I) how the change in the concentration of the reactant or product is being monitored per unit time, and

.....
.....
.....
.....

(II) how the concentration of the reactants are varied such that the order with respect to each reactant can be determined

.....
.....
.....
.....
.....
.....

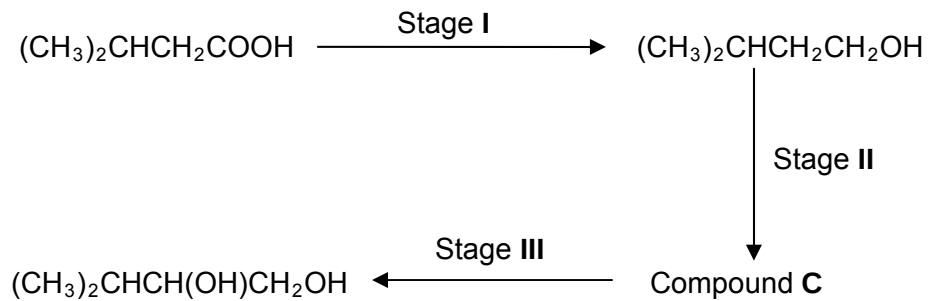
[4]

(iii) When **B** is oxidized, it produces carboxylic acid $(\text{CH}_3)_2\text{CHCH}_2\text{COOH}$. The acid has a boiling point of 176 °C while 3-methylbutylethanoate has a boiling point of 142 °C. Account for the relative values of their boiling points.

.....
.....
.....
.....
.....

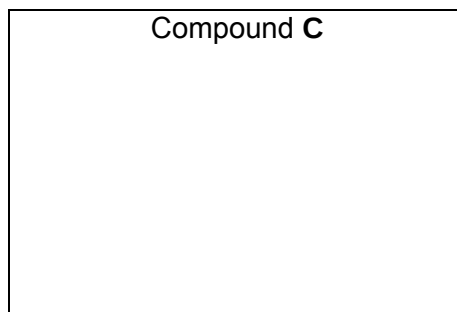
[2]

- 4 (b) (iv) The following shows a synthetic pathway starting from $(\text{CH}_3)_2\text{CHCH}_2\text{COOH}$.



Fill in the table below and give an appropriate structure for compound C.

	Reagents and Conditions
Stage I	
Stage II	
Stage III	



[4]
[Total: 17]